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Introduction

Artificial General Intelligence (AGI) is a hypothetical form of artificial intelligence designed to learn, reason, adapt, and apply knowledge across a wide range of tasks and domains at a level comparable to human intelligence (Bergmann & Downie, 2026). Unlike many of today's AI systems, which are designed or optimized to perform specific tasks, AGI would theoretically be able to transfer knowledge between different situations, solve unfamiliar problems, and adapt as new information becomes available.



This case study explores how AGI could help address that challenge through a proposed **Lifelong Workforce Intelligence System**. Rather than simply recommending a course or matching a person to a job, the system would consider an individual's capabilities, personal interests and desired lifestyle, changes

in the labor market, learning and employment opportunities, and potential career pathways.

By reasoning across these areas together, AGI could create a more personalized and adaptable approach to education and workforce development while helping individuals make better-informed decisions about what they should learn and where they should go next.

Industry Analysis

Education and workforce development play a critical role in preparing people for employment, career advancement, and economic mobility. However, the rapid growth of artificial intelligence is creating a challenge for both systems because the skills required in the workplace are changing faster than traditional education and training models can always respond. This does not mean that colleges and workforce development programs are failing completely. Instead, there is a growing need for these systems to become more adaptable and better connected to what is happening in the workforce.



One of the clearest challenges is the growing gap between the skills employers need and the skills workers are developing.

According to the National Association of Colleges and Employers (NACE), more than one-third of entry-level jobs surveyed in 2026 required AI skills, nearly three times the

percentage reported only six months earlier. Additionally, 28% of employers reported specifically seeking early-career employees who could effectively use AI in their work.

However, 50.5% of graduating seniors surveyed said they were not actively developing AI skills, while nearly one-third believed AI skills would have little or no importance to their future careers (National Association of Colleges and Employers, 2026). This disconnect highlights how quickly employer expectations are changing compared to how some students are preparing for the workforce.

The challenge extends beyond students entering their first careers. Existing workers are also being asked to adapt as AI changes workplace processes and responsibilities. PwC's *Global Workforce Hopes and Fears Survey 2025* found that only 51% of non-management employees believed they had access to the learning and development resources they needed, compared with 72% of senior executives (PwC, 2025). Similarly, Pew Research Center found that 52% of U.S. workers expressed concern about the future use of AI in the workplace, while 33% reported feeling overwhelmed by the pace of change (Pew Research Center, 2025). These findings suggest that preparing the workforce for AI requires much more than simply providing access to new technology. Workers also need guidance on how their roles are changing, which skills are becoming more valuable, and how emerging technologies can be integrated into their everyday work.

Artificial intelligence is already beginning to address parts of this challenge. Higher education institutions are exploring AI across learning, assessment, student advising, academic support, and administrative functions to provide more personalized educational experiences (Gamby et al., 2026). Workforce platforms are also using AI to identify employee skills and recommend career opportunities. For example, Workday Skills Cloud uses AI to analyze workforce capabilities and identify skill gaps, while LinkedIn's Next Role Explorer combines labor-market data, professional profiles, and AI to suggest potential career transitions (LinkedIn, n.d.; Workday, n.d.).

This fragmentation presents an opportunity for a fundamentally different approach. Rather than relying on multiple independent AI systems, an AGI-powered solution could continuously reason across education, workforce trends, personal goals, and learning opportunities simultaneously. Instead of simply reacting to changes in the labor market, education and workforce development could become more proactive, personalized, and connected to the future of work.

AGI Application Proposal: Lifelong Workforce Intelligence System

To address the growing disconnect between education, workforce development, and the rapidly changing labor market, this case study proposes a theoretical **Lifelong Workforce Intelligence System** powered by Artificial General Intelligence. The purpose of the system would be to help

individuals better understand where they are professionally, where the workforce is heading, and what they should learn or experience next. Rather than focusing only on degrees, certifications, or job titles, the system would continuously develop a holistic understanding of the individual



and use that knowledge to generate personalized education and career pathways.

The system would operate across **five connected pillars**, with AGI reasoning across all of them simultaneously.

Capabilities and Experience would establish what an individual can do by analyzing

education, certifications, work experience, demonstrated skills, projects, and other verified professional achievements using user-approved educational and employment data.

Interests, Identity, and Life Design would focus on the individual beyond professional credentials by considering personal interests, career goals, preferred work style, lifestyle priorities, and other information the user voluntarily chooses to share. Optional connections to selected digital platforms could provide additional context but would serve only as supporting signals rather than factual conclusions.

World of Work would continuously monitor labor-market trends, employer demand, emerging technologies, evolving occupations, compensation trends, and automation to understand how industries are changing over time.

The Learning and Opportunity Ecosystem would connect individuals with degree programs, certifications, apprenticeships, employer training, mentorship, entrepreneurship resources, professional organizations, and current job opportunities that align with their goals.

Pathway Intelligence would combine information from the previous four pillars to generate multiple evidence-based career pathways, identify skill gaps, recommend learning opportunities, and provide confidence windows that help individuals adapt as workforce conditions evolve.

Artificial General Intelligence would serve as the intelligence operating across all five pillars. Current AI systems are capable of performing individual tasks such as recommending courses, identifying skills, matching candidates to jobs, or generating career advice. However, these systems typically operate within narrowly defined objectives and depend heavily on the information explicitly provided to them. AGI, in contrast, could theoretically reason across multiple domains simultaneously, transfer knowledge between them, adapt to changing circumstances, and recognize relationships that specialized AI systems may overlook.

For example, an experienced accountant interested in AI but unwilling to pursue another degree may discover opportunities in AI governance, intelligent business operations, or financial automation consulting. Rather than recommending a single career, the system would present multiple pathways, explain why each aligns with the individual's goals, and identify the most efficient way to close any remaining skill gaps.

Each recommendation would include a confidence window indicating how long current labor-market conditions suggest the pathway is likely to remain strong. Rather than encouraging constant career changes, the system would notify users only when meaningful shifts in technology or workforce demand justify reconsidering their path.

The benefits would extend beyond individuals. Colleges, employers, and workforce organizations could use aggregated, privacy-protected insights to identify skill gaps, improve training, and better align educational programs with evolving workforce needs.

Risks and Ethical Considerations

Although a Lifelong Workforce Intelligence System could provide significant benefits, the level of personalization required would also create important ethical concerns. One of the most significant would be data privacy. The system could potentially have access to educational records, employment history, professional profiles, assessments, certifications, work products, and other personal information. Users could also voluntarily connect professional, social,

lifestyle, or wellness platforms to help the system better understand their interests and preferences. While this additional information could improve personalization, it could also increase the amount of sensitive data being collected. Individuals would therefore need complete control over which sources are connected, how their information is used, and the ability to disconnect any source or delete their information at any time.

Bias presents another important challenge. If AGI learns from historical hiring decisions, workforce trends, salary information, or career progression data, it could unintentionally reinforce existing inequalities. Without careful oversight, recommendations could reflect historical biases rather than future opportunities. To reduce this risk, the system would require continuous bias testing, diverse training data, transparent decision-making processes, and meaningful human oversight.

Another concern is overreliance on AI. Recommendations from the system could influence major life decisions, including career changes, educational investments, relocation, or entrepreneurship. Because these decisions may affect someone's livelihood for years, the system should never present recommendations as guaranteed outcomes. Instead, it should provide multiple evidence-based pathways, confidence levels, explanations of the data supporting each recommendation, and the assumptions that influenced its reasoning.

Finally, accessibility must also be considered. If advanced AGI-powered career guidance were only available through expensive subscription services, it could increase existing inequalities between those who have access to sophisticated career resources and those who do not. Partnerships with colleges, workforce development organizations, employers, and public agencies could help make these capabilities available to a broader population.

Ultimately, the Lifelong Workforce Intelligence System should function as a decision-support tool rather than an authority over someone's future. Its purpose is not to determine who someone should become, but to provide better information, greater visibility into opportunities, and more personalized guidance while allowing individuals to make their own decisions.

Conclusion

Although the possibilities of this system are significant, the responsibilities may be even greater. A platform capable of understanding someone's professional history, interests, goals, behaviors, and desired lifestyle would require an extraordinary level of trust. Inaccurate information, privacy failures, bias, or poor reasoning could influence life-changing decisions and undermine confidence in the system itself.

For this reason, any future AGI platform of this kind would need to be built on a foundation of transparency, privacy, user control, explainability, and trust. Individuals should always understand why recommendations are being made, what information influenced those recommendations, and retain the ability to reject or ignore them.



AGI should never become the final authority on what someone should learn, where they should work, or who they should become. Instead, it should serve as a trusted guide that expands possibilities while leaving the final decisions in human hands.

The Lifelong Workforce Intelligence System is not intended to be the definitive solution to the challenges facing education and workforce development. Rather, it represents one theoretical example of how AGI could create a more connected, adaptive, and personalized learning ecosystem. Ultimately, the greatest value of AGI may not be its ability to predict someone's future, but its ability to help people discover opportunities they may never have considered on their own.

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